

Tiago Pateiro

Senior Software Engineer — Backend & AI Systems

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SUMMARY

Backend-first engineer, 13+ years across product and consultancy. Currently on PagerDuty's incubation team, building an agentic incident-response product that runs in production — LangGraph orchestration over Python services, streaming APIs, agent memory. Before that I was project lead on several Status Pages workstreams, taking a newly formed team from first commit to GA in six months, then through enterprise upsell. Co-inventor on three PagerDuty patent filings.

Design doc before code, and ADRs for the decisions worth revisiting later. I lead workstreams across team boundaries, mentor the engineers on them, and stay on a feature through launch and customer rollout rather than handing it off at merge.

Core stack: Python • .NET/C# • Elixir • Node.js • TypeScript • LangGraph • AWS • Terraform

EXPERIENCE

PagerDuty

Senior Software Engineer

CSOps Status team, then PD Advance / LABS (incubation).

Lisbon, Portugal

June 2022 – Present

AI incident response (2024 – present)

Agentic incident-response product: LangGraph orchestration over Python services, an SSE streaming layer, a Node BFF, and a Vite single-page client.

- Designed and led the metering service behind AI Actions, the usage-based consumption model for PagerDuty Advance. Every metered call passes through it: over 2,000 requests per minute at roughly 50ms, so usage accounting stays accurate and per-account limits hold without latency users can feel.
- Built the shared memory layer agents write their context into — designed the contracts and use cases directly with the other teams whose agents would depend on it, rather than shipping an interface and asking them to adapt. Now backs three agents in production at roughly 500 requests per minute inside a 120ms budget.
- Separated durable from transient state in the LangGraph checkpointer, so resumed runs replay persisted state without re-emitting intermediate tool traffic — removing a recurring class of context-overflow failures.
- Worked on Agent Connectors, which link the SRE Agent to a customer's observability and knowledge sources over API or MCP so it can query their existing tooling.
- Shipped the Skills mechanism for the SRE Agent: custom instructions and runbooks the agent loads at run time, so it follows a team's own procedures instead of generic ones. Presented internally at PagerDuty's AI showcase.
- Currently prototyping deeper-reasoning agents for diagnosis and remediation, to shift investigation load off whoever is on call.

Status Pages (2022 – 2024)

- Project lead for subscriptions and notifications — two of the three critical user journeys, and the flow the product's pricing is built on. Designed the notification service to be modular so channels could land incrementally, then shipped email, Slack, webhooks and SMS across three consecutive quarterly releases, each on its committed date.
- Led the storage migration from S3 to DynamoDB once the original design's performance assumptions stopped holding under growth: modelled the schema, wrote the ADRs, and delivered the provisioning. Read latency dropped from roughly 200ms to under 10ms.
- Built bring-your-own-identity-provider support for Private Status Pages, the premium enterprise tier: customers gate their page behind their own OIDC provider — Azure, Okta, PingID, Salesforce, JumpCloud — instead of exposing it publicly, with onboarding kept to minimal setup. Session handling and JWKS/openid-configuration caching served provider metadata from CloudFront to cut round-trips out to customer IdPs. Adoption reached roughly 48% of accounts; I ran the setup calls with enterprise customers directly and wrote the provider configuration guides into the company knowledge base.
- Co-led custom domain provisioning — DNS and email white-labelling so notifications send from the customer's own domain. Diagnosed timeouts in de-provisioning and moved the flow async via an ADR, without moving the release date.
- Co-authored the product design document, led the Kubernetes migration, and bootstrapped the codebase for a newly formed team. Ran the notification build-versus-buy analysis against the internal platform the team ultimately adopted.
- Owned observability: SLIs and SLOs behind the team's SLAs, structured JSON logging with request correlation, and a monitoring overhaul that cut on-call alert fatigue. Migrated the service to TypeScript and raised test coverage past 70%.

Stack: Python, Elixir, Node.js/TypeScript, LangGraph, AWS (Lambda, SQS, DynamoDB, CloudFront), Kubernetes, Terraform, MySQL, Redis, Elasticsearch

Defined.ai (formerly DefinedCrowd)

Senior Software Engineer → Lead

Speech and NLP training-data platform built on a global crowd workforce.

Lisbon, Portugal

July 2019 – May 2022

- Led a cross-functional team of eight — four backend, two front-end, two QA, working alongside a PM — building automation and data tooling that replaced manual crowd workflows, cutting human handling per task and improving margin on delivered datasets.
- Owned the technical design for human-in-the-loop quality systems, contributor e-learning and gamification. My scope was the pre-screening tests that gate expert-level work: a contributor had to pass a language-specific assessment — an en-US test before translation tasks, for instance — before that work was offered to them. Raised task throughput by 60%.

Stack: .NET, React, TypeScript, Azure, SQL Server, Spark, Kafka, Superset

Truphone

Software Engineer

Global eSIM provider and mobile network operator.

Lisbon, Portugal

August 2017 – June 2019

- Gathered stakeholder requirements and defined the architecture for customer-facing portals and mobile apps, on the product line responsible for over 60% of company revenue.
- Part of the team that shipped one of the first iPhone apps built on Apple's initial eSIM release — over-the-air data plans and profile provisioning.

Stack: .NET, React Native, TypeScript, MongoDB, Elasticsearch, SQL, Java, Spark

BookedBy

Software Engineer

Booking and management SaaS for multi-location service businesses.

Lisbon, Portugal

January 2016 – August 2017

- Shipped an online booking product that ran standalone or embedded in client sites, removing the need for per-client custom development.
- Built a central management platform pushing promotions, inventory and users across multi-store chains, plus the data warehouse behind manager reporting.

Stack: Node.js, React, Backbone.js, .NET, PostgreSQL, Docker, AWS

EARLIER EXPERIENCE

ROFF — Developer (2012–2016). Designed an API integrating SAP QM with touchscreen shop-floor workflows and led technical direction for SAP integrations. Warehouse management with path-optimisation; automated production cost allocation for manufacturing clients. .NET, SQL, SAP ABAP.

InspirennovIT — Developer (2011–2012). PDF digital-signature workflow using the Portuguese citizen card, with SharePoint approval routing. .NET, SQL Server.

FCT-NOVA — Research Grant Holder (concurrent, 2011–2012). Academic data warehouse for candidate analytics; extraction tooling and geographic clustering. Java, Pentaho BI, MySQL.

PATENTS

Co-inventor on three patent filings at PagerDuty.

TECHNICAL SKILLS

AI systems LangGraph, agent orchestration, tool calling, MCP, agent state and memory, prompt and context design, knowledge graphs

Programming Python, .NET/C#, Elixir, Node.js, TypeScript/JavaScript, SQL

Backend Distributed services, event-driven architecture, streaming APIs (SSE), OIDC/OAuth, design docs and ADRs

Cloud & infra AWS, Terraform, Kubernetes, Docker, CI/CD, observability (SLIs/SLOs, structured logging, dashboards)

Data MySQL, PostgreSQL, DynamoDB, Redis, Elasticsearch, Kafka, Spark

Languages Portuguese (native), English (fluent)

EDUCATION

MSc Knowledge Management & Information Systems — ISCTE-IUL, 2018

Postgraduate Diploma Computer Science — FCT-NOVA, 2012

BSc Computer Science — FCT-NOVA, 2010